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# Kelompok 10 Cyber



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# Deklarasi Tech Stack & Replikasi

Backend autodosing nutrisi AB Mix melon — teknologi asli dipetakan ke replika lab.

| Komponen | Proyek IoT asli | Target Node replika |
|---|---|---|
| **Sistem operasi** | Firebase cloud terkelola Google;<br>OS tidak dikelola kelompok | Ubuntu Server<br>SRV-IOT-KEL10 · 192.168.10.5 |
| **Penyimpanan data** | Firebase Realtime Database | Firebase Realtime Database Emulator<br>127.0.0.1:9000 |
| **Komunikasi** | HTTPS REST API · JSON | HTTP REST API · JSON (belum TLS)<br>Simulator Python → data dummy |
| **Dashboard** | Aplikasi mobile Flutter<br>terhubung ke Firebase | Web HTML/JavaScript melalui Nginx<br>/dashboard.html dan /api/sensor |

**BATAS SIMULASI Data sintetis saja; tidak menguji Firebase produksi, sensor fisik, atau pompa.**

SUMBER tech_stack_declaration.md · Panduan PBL Minggu 9, Bagian III–IV

# Arsitektur Simulasi & Skema IP

VirtualBox Host-Only · 192.168.10.0/24 · alamat statis · pengujian hanya antar-VM lab.

**ATTACKER / CLIENT**

**CyberOps Workstation**

`192.168.10.100/24`

SSH :22

HTTP :80

**TARGET · Ubuntu Server**

`192.168.10.5/24`

Nginx :80 → dashboard + API baca

`Firebase Emulator 127.0.0.1:9000`

Simulator Python → REST lokal

Port database tidak dibuka ke jaringan.

**HOST / ADMIN**

**Laptop pengguna**

`192.168.10.1/24`

**MONITORING · Security Onion**

eth1: 192.168.10.200/24 (manajemen)

eth0: sensor tanpa IPv4; Allow All

Sguil / tcpdump mengamati trafik lab.

Sensor mengamati trafik Host-Only, bukan jalur proxy.

NAT dinonaktifkan setelah instalasi; UFW bukan isolasi internet.

Trafik loopback :9000 tidak terlihat oleh sensor jaringan.

SUMBER ../design/ip_plan.md · diagram diperbarui mengikuti konfigurasi terdokumentasi

# Progres Hardening: Selesai & Belum

8 selesai (A:2 · B:3 · C:3) + 9 rencana belum terverifikasi · status mengikuti bukti, bukan klaim aman produksi.

| ID | Kontrol | Verifikasi / perubahan | Status |
|---|---|---|---|
| **A1** | Tolak SSH root langsung | permitrootlogin no; login root ditolak | Selesai |
| **A2** | Fail2ban untuk SSH | IP .100 diblokir; koneksi baru ditolak | Selesai |
| **B1** | UFW whitelist IP & port | :22/:80 open; :9000 open → filtered | Selesai |
| **B2** | Database hanya localhost | Listener jaringan → 127.0.0.1:9000 | Selesai |
| **B3** | Sembunyikan versi Nginx | nginx/1.18.0 (Ubuntu) → nginx | Selesai |
| **C1** | Rules & validasi payload | Data valid 200; humidity 150 → 401 | Selesai |
| **C2** | Backup & uji pemulihan | Backup 72 → diubah 80 → pulih 72 | Selesai |
| **C3** | Batasi permission aplikasi | Direktori 700; file 600; nobody ditolak | Selesai |

## BELUM / RENCANA LANJUTAN

- A3 SSH key-only
- A4 Pisahkan akun admin & layanan
- B4 Inventaris versi & patch paket
- B5 HTTPS dashboard / API
- C4 Akses berbasis identitas / peran
- C5 Review & rotasi log terpusat
- C6 Backup terjadwal di luar VM
- C7 Sinkronisasi waktu 3 VM
- C8 Batas ukuran & laju request API

Bukti before/after tersimpan per kontrol. A2 adalah Fail2ban, bukan SSH key-only.

Batas bukti: log impor backup dan pemulihan akses SSH belum didokumentasikan terpisah.

SUMBER ../blueteam/hardening_checklist.md · tautan screenshot before/after per kontrol

# Baseline Trafik Normal & Bukti Monitoring

CyberOps 192.168.10.100 → Ubuntu 192.168.10.5 · data dummy · pengamatan Sguil dan tcpdump.

### ICMP / UJI KONEKTIVITAS

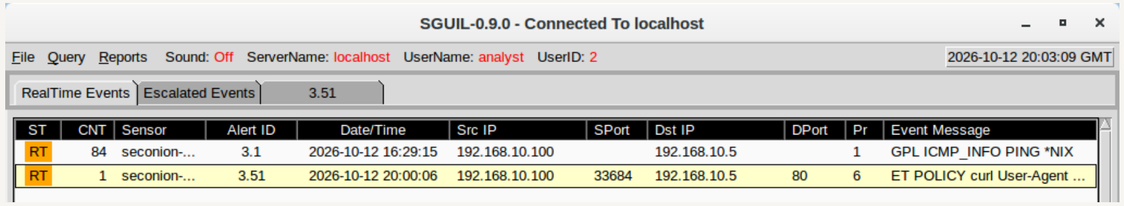
Ping mendapat balasan; Sguil mencatat

GPL ICMP_INFO PING *NIX dari .100 ke .5.

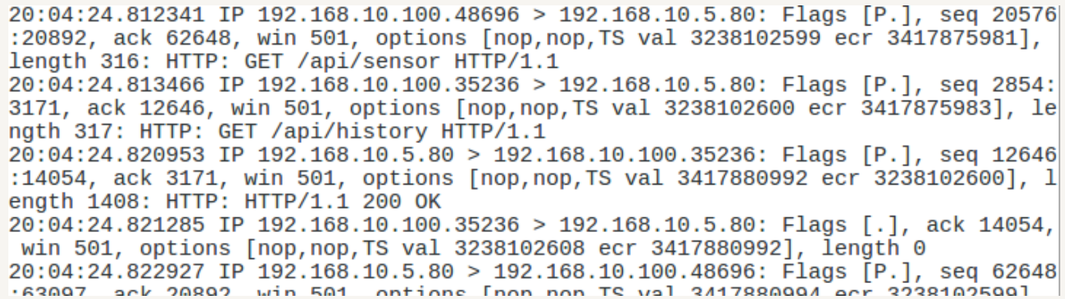
### HTTP :80 / PEMBACAAN API

GET /api/sensor dan /api/history → 200 OK.

Event kebijakan curl terlihat pada Sguil.




B03 · Sguil: event ping dan curl (cuplikan bukti).



```
20:04:24.812341 IP 192.168.10.100.48696 > 192.168.10.5.80: Flags [P.], seq 20576
:20892, ack 62648, win 501, options [nop,nop,TS val 3238102599 ecr 3417875981],
length 316: HTTP: GET /api/sensor HTTP/1.1
20:04:24.813466 IP 192.168.10.100.35236 > 192.168.10.5.80: Flags [P.], seq 2854:
3171, ack 12646, win 501, options [nop,nop,TS val 3238102600 ecr 3417875983], le
ngth 317: HTTP: GET /api/history HTTP/1.1
20:04:24.820953 IP 192.168.10.5.80 > 192.168.10.100.35236: Flags [P.], seq 12646
:14054, ack 3171, win 501, options [nop,nop,TS val 3417880992 ecr 3238102600], l
ength 1408: HTTP: HTTP/1.1 200 OK
20:04:24.821285 IP 192.168.10.100.35236 > 192.168.10.5.80: Flags [.], ack 14054,
 win 501, options [nop,nop,TS val 3238102608 ecr 3417880992], length 0
20:04:24.822927 IP 192.168.10.5.80 > 192.168.10.100.48696: Flags [P.], seq 62648
```

B06 · tcpdump: request API dan respons HTTP 200.

INTERPRETASI Event ping/curl bukan otomatis serangan. Interval ±5 detik masih rancangan, bukan statistik terukur.

Loopback :9000 tidak tercakup; temuan port 9000 pada bukti lama perlu diverifikasi ulang sebelum demo.

SUMBER ../blueteam/baseline_traffic.md · screenshot dapat diklik untuk membuka bukti utuh

# Live Demo Singkat: Security Onion

Target demo ±2–3 menit · buktikan visibilitas trafik normal, bukan serangan sungguhan.

## URUTAN DEMO DI LAYAR

01 Security Onion: cek sudo so-status; buka Sguil.

02 CyberOps: kirim ping dan GET ke API Target.

```
ping -c 4 192.168.10.5
curl -i http://192.168.10.5/api/sensor
```

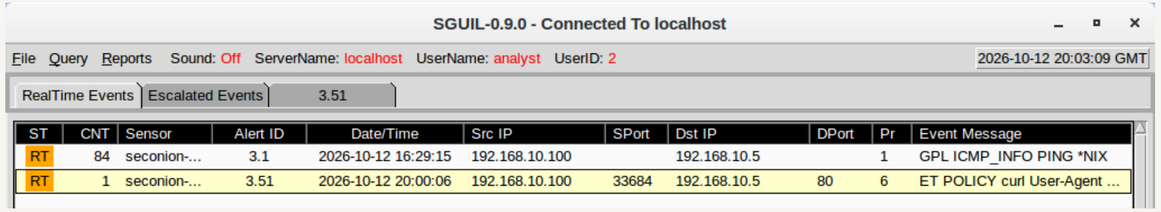
03 Sguil: cocokkan waktu, IP .100 → .5, dan event.

04 Konfirmasi ping reply + HTTP 200 / JSON.

Jika event tidak muncul, cek paket via tcpdump;

catat apa adanya, jangan klaim sudah terdeteksi.

## BUKTI PENDUKUNG / CADANGAN




Screenshot terdokumentasi — bukan hasil live sesi ini.

## SEBELUM DEMO

Host-Only aktif; NAT mati; sensor Allow All.

Selaraskan waktu VM dan verifikasi port 9000.

Data dummy saja; tanpa scanning ke sistem mitra.

SUMBER ../blueteam/baseline_traffic.md · runbook pada catatan pembicara slide ini

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# Phishing Attacks

Phishing tricks users into giving sensitive information through fake messages or websites

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# Ransomware

Ransomware locks data and demands payment to restore access

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# Cyber Security in Daily Life



## Online banking



## Social media accounts



## Cloud storage

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# Cyber Security for Business

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# Basic Security Practices



- **Strong passwords**
- **Regular updates**
- **Secure networks**

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# Tools Used in Cyber Security

## Firewalls

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## Antivirus software

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## Encryption

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# Careers in Cyber Security

- Security Analyst
- Ethical Hacker
- Network Security Engineer



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# Conclusion

Cyber security is essential to keep data safe and ensure digital trust

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# Thank You

Thank you for your attention


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